

The Surat Technical Education & Research Society
Smt. Kalavatiben Fulchandbhai Vakharia Education Complex
SASCMA English Medium Commerce College &
Shri Hasmukhbhai Hojiwala College of Business Administration (BBA) &
Smt. Ushaben Jayvadan Bodawala College of Computer Application (BCA)
Affiliated to Veer Narmad South Gujarat University

A
SEMINAR REPORT
ON
“ Air Quality Monitoring Using Drone Technology ”

SUBMITTED AS A PARTIAL REQUIREMENT
FOR BACHELOR OF COMPUTER
APPLICATION
(SEMESTER VI)



GUIDED BY:
Bhumi Patel

SUBMITTED BY:
Chothiyawala Tanishq V. (5946)

YEAR 2024-25

ACKNOWLEDGEMENT

When we reach on completion of the project, giving credit becomes a must, without the support of so many people's help and guidance this project couldn't be completed successfully.

First, we would like to thank our parents with their great support we can reach at the stage. Then our humble thanks to all professors and all staff members of our college **SASCMA BCA COLLEGE** , For the Co-operation & interest extended by them, throughout our degree course. It is a base that they have built, which sustains such good jobs.

We would like to thank **Dr. Ashish Desai** I/C Principal of **SASCMA BCA COLLEGE** for granting us an opportunity to work on this project. We would also like to thank to our guide **Bhumi Patel** for their valued guidance and encouragement towards the completion of our project.

INDEX

Sr no.	Description	Page No.
1	Introduction	1
2	Overview of Drone Technology	2
3	Overview of Air Quality Monitoring System	11
4	Previous Technology for Air Quality Monitoring System	14
5	How Air Quality Monitoring System Overcomes Previous Technology	16
6	Hardware and Software Components	18
7	How it works (Operation and working Mechanism)	20
8	Analysis	24
9	Future Enhancements	26
10	Conclusion	29
11	Reference	30

Abstract

This seminar documentation explores **Air Quality Monitoring Using Drone Technology** by examining its key principles, methodologies, and real-world applications. The study delves into **sensor integration, real-time data transmission, GIS mapping, and AI-based pollution analysis**, highlighting their significance in **environmental science, public health, and smart city development**. Through an in-depth analysis, the documentation presents **the effectiveness of drone-based monitoring in detecting pollutants, improving response time, and providing high-resolution air quality data**, emphasizing the impact and potential advancements in the subject. Furthermore, challenges, solutions, and future scope are discussed to provide a comprehensive understanding of the topic. This work serves as a valuable resource for researchers, students, and professionals seeking insights into **Air Quality Monitoring Using Drone Technology**.